



**Cambridge Instruments  
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

OIL TEST SET · TRANSFORMER OIL BDV TESTER

# CIE/7100

Transformer oil dielectric strength (BDV) tester up to 100 kV.

UP TO 100KV

## Transformer Oil BDV Tester

Part of the CIE/7000 Series oil dielectric strength test set range, built for breakdown/withstand voltage testing of transformer and circuit breaker insulating oil per IS 6792:1992 – a cast-resin H.V. transformer with graded insulation, zero-start interlocking, and an interlocked H.T. door cutout for safe operation.

- **0–100kV output** – continuously variable, 1000 VA capacity
- **Manual or motorised** – motorised advances at approx. 2kV/second
- **Memory device** – retains breakdown reading after tripping
- **Zero-start interlocking** – plus interlocked H.T. door cutout
- **Analog moving-iron metering** – 96mm<sup>2</sup> kV meter, L.T. side
- **Standard test vessel** – brass spheres, per IS 6792:1992



- V Up to 100kV** Continuously variable, 1000 VA
- M Manual/Motorised** Approx. 2kV/second (motorised)
- ✓ Memory Device** Retains breakdown reading

*Manufactured in India since 1963 – trusted by various State Electricity Boards, Indian Railways, cement and metal industries, and other sectors.*

### 01 Full Specification Sheet

| PARAMETER             | SPECIFICATION                                                                   |
|-----------------------|---------------------------------------------------------------------------------|
| <b>Output Voltage</b> | 0–100 kV (continuously variable)                                                |
| <b>Capacity</b>       | 1000 VA                                                                         |
| <b>Input</b>          | 230 V ±10%, 50 Hz, 1 phase A.C.                                                 |
| <b>Duty</b>           | Intermittent                                                                    |
| <b>H.V. Winding</b>   | Cast-resin, graded insulation, centre point grounded                            |
| <b>Metering</b>       | 96 mm <sup>2</sup> A.C. moving-iron voltmeter, L.T. side, scaled in kV (analog) |
| <b>Motorised Rate</b> | Approx. 2 kV/second (motorised variant)                                         |
| <b>Test Vessel</b>    | Two brass spheres, 12.5–13 mm dia., 4mm or 2.5mm gap, per IS 6792:1992          |
| <b>Warranty</b>       | 12 months against manufacturing defects                                         |

### 02 CIE/7000 Series - Output Voltage & Capacity

| OUTPUT VOLTAGE | CAPACITY | MANUAL MODEL | MOTORISED MODEL |
|----------------|----------|--------------|-----------------|
| 0–50 kV        | 500 VA   | CIE/7050     | CIE/7050 MOT    |
| 0–60 kV        | 600 VA   | CIE/7060     | CIE/7060 MOT    |
| 0–70 kV        | 700 VA   | CIE/7070     | CIE/7070 MOT    |
| 0–75 kV        | 750 VA   | CIE/7075     | CIE/7075 MOT    |
| 0–80 kV        | 800 VA   | CIE/7080     | CIE/7080 MOT    |
| 0–90 kV        | 900 VA   | CIE/7090     | CIE/7090 MOT    |
| 0–100 kV       | 1000 VA  | CIE/7100     | CIE/7100 MOT    |
| 0–120 kV       | 1200 VA  | CIE/7120     | CIE/7120 MOT    |

### 03 Typical Applications

- Transformer oil dielectric strength testing before energisation
- Routine oil quality monitoring for distribution and power transformers
- Circuit breaker oil testing
- Power utility and substation maintenance laboratories

### 04 What's in the Box

- ✓ Instruction manual with circuit diagram
- ✓ Input supply cable, minimum 2 metres, 3-pin plug
- ✓ Test vessel (oil cell) with electrode and gap-setting gauge

### 05 Optional Accessories

- + Alternate tripping current set point
- + Digital metering in place of analog
- + 2-unit housing
- + Magnetic stirrer